

Recommended Course Sequence
Aerospace Engineering Option B – Structures and Materials (September Entry)
2026-2027 Academic Year

	SUMMER /1		FALL /2		WINTER /4	
YEAR 1			AERO 201 Intro to Flight & Aero Systems (4.00) The following course must be completed previously or concurrently: ENGR 213.		AERO 253 Thermal Sciences for Aerospace (3.00) The following course must be completed previously: MATH 205; ENGR 213.	
			ENGR 213 Applied Ord. Differential Eq. (3.00) The following course must be completed previously or concurrently: MATH 204 (Cegep Mathematics 105). The following course must be completed previously: MATH 205 (Cegep Mathematics 203).		ENCS 282 Technical Writing & Comm. (3.00) Students must have satisfied the requirements in Section 71.20.7 Writing Skills Requirement, by passing the Engineering Writing Test (EWT) or by passing ENCS 272 with a grade of C- or higher, prior to enrolling.	
			ENGR 201 Professional Practice & Resp. (1.50) Prerequisites: none.		ENGR 233 Applied Advanced Calculus (3.00) The following course must be completed previously: MATH 204 (Cegep Mathematics 105); MATH 205 (Cegep Mathematics 203).	
			ENGR 242 Statics (3.00) The following course must be completed previously or concurrently: ENGR 213. The following courses must be completed previously: PHYS 204; MATH 204.		ENGR 243 Dynamics (3.00) The following courses must be completed previously: ENGR 213, ENGR 242.	
			MIAE 215 Programming for Mech & Indu Eng. (3.50) The following course must be completed previously: MATH 204 (Cegep mathematics 105).		ENGR 244 Mechanics of Materials (3.75) The following courses must be completed previously: ENGR 213; ENGR 242 or ENGR 245. The following courses must be completed previously or concurrently: ENGR 233.	
YEAR 2			ENGR 202 Sust. Dev. Enviro. Stewardship (1.50) Prerequisites: none.		AERO 290 Introduction to Aircraft Design (3.00) The following course must be completed previously: AERO 201. The following course must be completed previously or concurrently: ENCS 282.	
			ENGR 311 Transform Calc. & Partial Diff. Eq. (3.00) The following courses must be completed previously: ENGR 213, ENGR 233.		AERO 371 Modelling and Control Systems (3.50) The following courses must be completed previously: PHYS 205; ENGR 213, ENGR 243. The following course must be completed previously or concurrently: ENGR 311 or ELEC 342 or ELEC 364.	
			ENGR 371 Probability & Stats in Eng. (3.00) The following courses must be completed previously: ENGR 213, ENGR 233.		ENGR 361 Fluid Mechanics I (3.00) The following courses must be completed previously: ENGR 213, ENGR 233, ENGR 251 or AERO 253.	
			MIAE 211 Mech. Engineering Drawing (3.50) Prerequisites: none.		MECH 343 Theory of Machines (3.50) The following courses must be completed previously: ENGR 213, ENGR 233, ENGR 243.	
			MIAE 221 Materials Science (3.00) The following course must be completed previously: CHEM 205 (Cegep Chemistry 101).		MIAE 313 Machine Drawing and Design (3.50) The following course must be completed previously: MECH 211 or MIAE 211.	
YEAR 3			AERO 390 Aerospace Engr. Design Project (3.00) The following courses must be completed previously: AERO 290, AERO 371; ENCS 282.		ENGR 301 Engr. Manage. Principles Econ (3.00) Prerequisites: none.	
			AERO 417 Standards, Reg. and Certification (3.00) The following course must be completed previously: ENGR 201.		ENGR 391 Numerical Methods in Engr. (3.00) The following courses must be completed previously: ENGR 213, ENGR 233; COMP 248 or COEN 243 or MECH 215 or MIAE 215 or BCEE 231.	
			AERO 481 Materials Engr. for Aerospace (3.50) The following course must be completed previously: MECH 221 or MIAE 221.		ENGR 392 Impact of Technology on Society (3.00) The following courses must be completed previously: ENCS 282; ENGR 201, ENGR 202.	
			MECH 373 Instrumentation & Measurements (3.50) The following courses must be completed previously: ENGR 311; AERO 371 or MECH 370 or MECH 371.		MECH 375 Mechanical Vibrations (3.50) The following course must be completed previously: AERO 371 or MECH 370 or MECH 371.	
			MIAE 311 Manufacturing Processes (3.00) The following course must be completed previously: MECH 313 or MIAE 313.		MIAE 383 Applied Machine Learning MIAE (3.50) The following courses must be completed previously: ENGR 371; MIAE 215.	
			MIAE 312 EDML Lab (1.00) The following course must be completed previously or concurrently: MIAE 311.			
YEAR 4			AERO 431 Principles of Aeroelasticity (3.50) The following courses must be completed previously: ENGR 361; MECH 375.		AERO 487 Design of Aircraft Structures (3.00) The following course must be completed previously: AERO 486.	
			AERO 486 Aircraft Stress Analysis (3.00) The following courses must be completed previously: ENGR 243, ENGR 244.		MECH 460 Finite Element Analysis (3.75) The following courses must be completed previously: ENGR 244, ENGR 391.	
			MECH 412 Computer-Aided Mech. Design (3.50) The following course must be completed previously: MECH 311 or MIAE 311.		General Studies (3.00) (Undergraduate Calendar, Sec. 71.110)	
			AERO 490 Capstone Aerospace Engineering Design Project (6.00) The following courses must be completed in advance: AERO 390; ENGR 301. Students must have completed 75 credits in the program prior to enrolling.			

DETAILED COURSE INFORMATION
Aerospace - Option B 2026-27

COURSE	TITLE	CREDIT	PRE-REQUISITE	CO-REQUISITE	SUM 1	SUM 2	FALL	WIN
AERO 201	Introduction to Flight and Aerospace Systems	4.00	ENGR 213				X	X
AERO 253	Thermal Sciences for Aerospace Engineering	3.00	ENGR 213; MATH 205					X
AERO 290	Introduction to Aircraft Design	3.00	AERO 201	ENCS 282			X	X
AERO 371	Modelling and Control Systems	3.50	PHYS 205; ENGR 213, ENGR 243	ENGR 311 or ELEC 342 or ELEC 364			X	X
AERO 390	Aerospace Engineering Design Project	3.00	AERO 290, AERO 371; ENCS 282				X	X
AERO 417*	Standards, Regulations and Certification	3.00	ENGR 201		X*		X	
AERO 431	Principles of Aeroelasticity	3.50	ENGR 361; MECH 375				X	
AERO 481	Materials Engineering for Aerospace	3.50	MECH 221 or MIAE				X	
AERO 486	Aircraft Stress Analysis	3.00	ENGR 243, ENGR 244				X	
AERO 487	Design of Aircraft Structures	3.00	AERO 486					X
AERO 490	Capstone Aerospace Engineering Design Project	6.00	AERO 390; ENGR 301. Students must have completed 75 credits in the program.				X	
ENCS 282	Technical Writing and Communication	3.00	Passing the Engineering Writing Test (EWT) or ENCS 272 with a grade of C- or higher.		X	X	X	X
ENGR 201	Professional Practice and Responsibility	1.50			X		X	X
ENGR 202	Sustainable Development and Environmental Stewardship	1.50			X		X	X
ENGR 213	Applied Ordinary Differential Equations	3.00	MATH 205 (Cegep Mathematics 203)	MATH 204 (Cegep Mathematics 105)	X		X	X
ENGR 233	Applied Advanced Calculus	3.00	MATH 204 (Cegep Mathematics 105); MATH 205 (Cegep Mathematics 203)		X	X	X	X
ENGR 242	Statics	3.00	ENGR 213	PHYS 204; MATH 204	X		X	X
ENGR 243	Dynamics	3.00	ENGR 213, ENGR 242		X		X	X
ENGR 244	Mechanics of Materials	3.75	ENGR 213; ENGR 242 or ENGR 245	ENGR 233	X		X	X
ENGR 251	Thermodynamics I	3.00	MATH 203		X	X	X	X
ENGR 301	Engineering Management Principles and Economics	3.00			X	X	X	X
ENGR 311	Transform Calculus and Partial Differential Equations	3.00	ENGR 213, ENGR 233		X	X	X	X
ENGR 361	Fluid Mechanics I	3.00	ENGR 213, ENGR 233, ENGR 251 or AERO 253		X		X	X
ENGR 371	Probability and Statistics in Engineering	3.00	ENGR 213, ENGR 233		X	X	X	X
ENGR 391	Numerical Methods in Engineering	3.00	ENGR 213, ENGR 233; COMP 248 or COEN 243 or MECH 215 or MIAE 215 or BCEE 231			X	X	X
ENGR 392	Impact of Technology on Society	3.00	ENCS 282; ENGR 201, ENGR 202		X	X	X	X
Gen. Ed.	General Education Elective	3.00	See section 71.7110 of the Undergraduate Calendar		X	X	X	X
MECH 343	Theory of Machines	3.50	ENGR 213, ENGR 233, ENGR 243				X	X
MECH 352	Heat Transfer I	3.50	ENGR 311, ENGR 361				X	X
MECH 373	Instrumentation and Measurements	3.50	ENGR 311; AERO 371 or MECH 370 or MECH 371				X	
MECH 375	Mechanical Vibrations	3.50	AERO 371 or MECH 370 or MECH 371			X	X	X
MECH 412	Computer-Aided Mechanical Design	3.50	MIAE 311				X	
MECH 460	Finite Element Analysis	3.75	ENGR 244, ENGR 391					X
MIAE 211	Mechanical Engineering Drawing	3.50			X		X	X
MIAE 215	Programming for Mechanical, Industrial and Aerospace Eng	3.50	MATH 204 (Cegep mathematics 105)			X	X	X
MIAE 221	Materials Science	3.00	CHEM 205 (Cegep Chemistry 101)				X	X
MIAE 311	Manufacturing Processes	3.00	MECH 313 or MIAE 313		X		X	
MIAE 312**	Engineering Design and Manufacturing Processes Lab	1.00		MIAE 311	X**		X	
MIAE 313	Machine Drawing and Design	3.50	MECH 211 or MIAE 211			X	X	X
MIAE 383	Machine Learning for MIAE	3.50	ENGR 371; MIAE 215 or COEN 243				X	X

*AERO 417 reserved for AERO students in summer
**MIAE 312 reserved for Co-op students in summer

Note: In the case of discrepancies between this and the current Undergraduate Calendar, please contact your Undergraduate Program Assistant for clarification.
This information was compiled April 2026.

Registration Essentials

How do I register for classes?

Student registration is accessible within the [Student Centre](#) or the [My CU Account](#) page of the Student Hub. Detailed information and how-to guides can be found here [Course registration - Concordia University](#).

If you need assistance for any particular reason, please email your [Undergraduate Program Assistant](#) and include your Student ID as well as detailed course information (i.e. term, lecture, lab and tutorial), as follows:

Information we need:	Where to find the information	Example:
Summer/Fall/Winter YEAR Lec ABCD 123-CC (2964) Tut ABCD 123-CCF (2969) Lab ABCD 123- CI-X (2966)	MIAE 215 Programming for Mechanical and Industrial Engineers LECT 7306 Seats: 62/90 LAB GI-X 7321 Seats: 21/24 TUT T TB 7308 Seats: 14/48 This is the course information that we need. ← Sample taken from the Visual Schedule Builder	Fall 2026 LEC MIAE 215 T (7306*) Tut MIAE 215 TB (7308*) Lab MIAE 215 GI-X (7321*) *Code is optional

Registration Regulation: The 'C-' Rule

All **200-level courses** within the program [including admission/ECP/MEP requirements] which are prerequisites for other courses must be completed with a C- or higher. A 200-level course in which a student has obtained a D+ or lower must be repeated before attempting a course for which it is a prerequisite (Undergraduate Calendar, [Section 71.10.4](#)). **This does not apply to 300 or 400 level courses.**

Registration Regulation: The '200-level before 400-level' Rule

Students must complete (with a posted final grade) all 200-level courses required for their program before registering for any 400-level courses (Undergraduate Calendar, [Section 71.10.4](#)).

Why can't I enroll for ENCS 282?

Before enrolling in ENCS 282, students must meet the Writing Skills Requirement (Undergraduate Calendar, [Section 71.20.7](#)) by either passing the Engineering Writing Test (EWT) or ENCS 272 with a grade of C or higher. **This is faculty policy, no exceptions.** More details about the EWT, including registration and contact info, can be found here: [Engineering Writing Test](#).

Where can I find the list of General Studies courses?

You can find the list of available courses here: [Section 71.110 Complementary Studies for Engineering and Computer Science Students](#). Please note the [Exclusion List](#), as these courses may not be taken to fulfill this requirement.

Do you need to re-sequence?

Please use [our website's](#) re-sequencing guide, planning tools, and [sequence template](#) to draft your course plan through to graduation. On page 2 of your recommended sequence, you'll find a table outlining all prerequisites, co-requisites, and course offerings by term—this will help with your planning. You can send a draft to your Program Assistant for review.

Registration: 3rd Time Repeat

A student may repeat a failed course only once. A student who fails a required course twice must request permission to take the course a third time. To submit a request, please go to: [GCS Student requests & forms](#)

How do I register for Capstone?

Online registration for Capstone opens June 1, after the annual assessment of GPA is performed by the Registrar's Office. For those taking prerequisite courses in either summer term, you can only register once the grades are available.

How do I know if I am on track for graduation?

The best way to assess what is required for your program is to look at your [Academic Requirements Report](#). The report outlines the courses and credits required for your program(s), including majors, minors, and electives. It helps you monitor completed and outstanding requirements, track total credits, and note any deficiencies or transfer credits. *This is a reference tool only—consult your program advisor for official guidance.*

All of the information above, and more, can be found on our [Quick Tips and FAQ \(MIAE\)](#)

For more information, or clarification, please do not hesitate to contact your [Undergraduate Program Assistant](#) or [Student Academic Services](#)